

complex and humanly inefficient activities (e.g., fourth umpire reviews in cricket) and, above all, generating historical data for future analysis and training.

Today, the sports industry is using data analytics in various ways.

Dynamic modelling handles changing conditions and prepares focused data models for quicker action, including match analysis, player analysis, venue analysis, and forecasting. This kind of dynamic modelling helps to build robust and reliable data services quickly.

Data cleaning and preprocessing helps to use the right set of data for analytics, data science, and predictive analysis. It enables cleaning up of unwanted data and eventually helps in accelerating data analysis due to the use of focused data sets.

Data management analyses historical data and predicts future trends using predictive analysis by managing data science and machine learning models.

In many sports events, Big Data analytics is used for predictive and historical data analysis using data science; IoT is used for data collection and transfer from the entire field area, cloud computing is used to prepare scalable and agile architecture; AI and ML are used for statistical analysis, model training, and anomaly behaviour detection (e.g., pose analysis, field condition analysis, player performance analysis); and drones are used to capture and collect data for data analytics.

In short, the use of technology in sports events enhances the user experience, reduces manual errors, and records various activities for training and analytics purposes.

Time series database solutions

When you plan for data analytics or metrics monitoring, you need data that is bound to time series. As an example, time bound data is used to

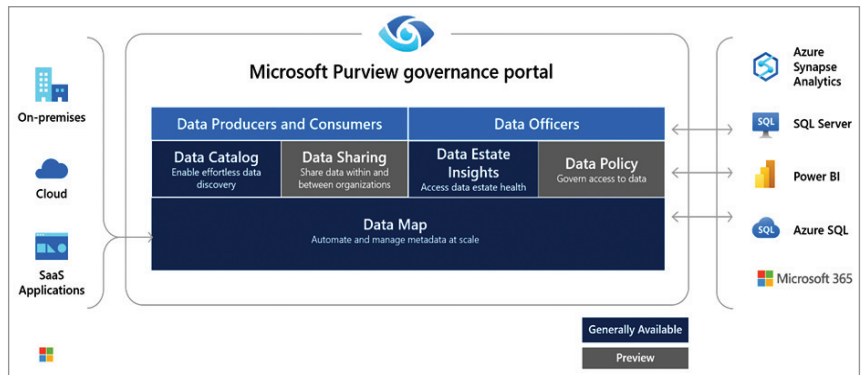


Fig. 1: Component view of Microsoft Purview

analyse server performance during a given period of time, or understand the market reach of a product for a given period of time to forecast future trends. Time bound data analysis helps to prepare real-time inferences based on facts.

Traditionally, time series data was handled by relational database management systems (RDBMS), but these were not found to be efficient for real-time analytics and statistical evaluation. The use of time series databases (TSDB) started in 2011, beginning with OpenTSDB, and got more popular with Facebook's Gorilla (in-memory data store) solution. Now time series databases are more popular for time series event analysis and data analytics.

Time series databases store data in a sequence of time intervals to track data changes over time. These purpose-built solutions have three features to handle time series data:

- Time stamp based recording of data in the database
- Time stamps that are associated with time zones and calendars automatically
- Sampling frequency such as periodic time sampling and discrete sampling

In cloud platforms, time series data storage is more popularly used for observability, IoT enabled services, and event analysis functionality. For example, Microsoft

introduced its time series database as Amazon Timestream in 2018.

Microsoft has introduced time series data visualisation with Azure Data Explorer, analytics using Azure time series insights (TSI), and time series metrics analytics with Google BigTable. Prometheus is the most popular third-party monitoring and management service across all cloud platforms—it is efficient in time series data visualisation and analysis.

OpenTSDB, InfluxDB, and Apache Druid are also popular time series database services popularly used for events, trend analysis, market trends, stock market changes, application performance monitoring, and infrastructure KPI analysis.

Time series databases are popular for banking and financial data analysis, IoT data collection and monitoring, stock and share trading, retail market trend analysis, market basket analysis, and health care trend analysis.

Microsoft Purview: Data management solution on the Azure platform

In 2021, Microsoft launched Azure Purview, now known as Microsoft Purview. This cloud native data governance solution helps organisations manage cloud and data on-premises.

Tons of features are packed into Microsoft Purview (Fig. 1):

- Data discovery to create a